

EVALUATING INTERDISCIPLINARY AND PROJECT-BASED LEARNING ON AI IN CULTURE AND ARTS

T. Sanchez¹, B. Zönnchen¹, H. Held², M. Dzhimova², G. Socher¹

¹*Hochschule München University of Applied Sciences - Munich Center for Digital Sciences and AI (GERMANY)*

²*University of Music and Performing Arts Munich (HMTM) (GERMANY)*

Abstract

In 2022, the Bavarian Research Institute for Digital Transformation (BIDT) called for digitalization labs, i.e., interdisciplinary teaching curricula among public universities. Digitalization labs promote inclusive and democratic participation in digitalization issues through interdisciplinarity, inquiry-based learning, and project-oriented work. In this context, "Artificial Intelligence in Culture and Arts" (AICA) is a digitalization lab founded in cooperation with Hochschule München University of Applied Sciences (HM) and the University of Music and Performing Arts Munich (HMTM). AICA aims to equip students with the knowledge and skills to comprehend and impact AI development in the creative and cultural industries. Little hindsight has been gained on digitalization labs to engage students from diverse expertise and fields in digital transformation issues. This work reflects on the efficacy of AICA, particularly its project-based course. Does this course foster AI literacy and self-efficacy in impacting the deployment of AI in culture and arts among students with contrasted technological prior expertise?

This article introduces the course and the five projects conducted by its participants. We then present a questionnaire administered to the 12 participants of the course. The questionnaire comprises two parts besides demographics. First, Likert-scale and open-ended questions investigate self-reported benefits in line with the digitalization lab objectives. Second, a multiple-choice question with a single correct answer investigates AI literacy from a social and technical standpoint. The results suggest that students perceive benefits of the project course regarding their self-efficacy, knowledge, and skills regarding AI in culture and arts. These benefits seem independent of prior expertise and primary field of study. This work also applies and reflects on an objective measurement of AI literacy. We discuss its results and propose 16 refined questions to be used in future investigations.

Keywords: Artificial intelligence, culture and art fields, interdisciplinary teaching, project-based learning.

1 INTRODUCTION

In 2022, the Bavarian Research Institute for Digital Transformation (BIDT) called for digitalization labs, a new 4-year funding format for interdisciplinary teaching and learning formats among Bavarian public universities. Teaching formats should promote interdisciplinarity, inquiry-based learning, and project-oriented work for students to comprehend and develop solutions addressing today's digital transformation challenges. Digitalization labs are led by established academics and actively supervised and shaped by one or two coaches.

The digitalization lab "AI in Culture and Arts" (AICA) was funded in cooperation with Hochschule München University of Applied Sciences (HM) and the University of Music and Performing Arts Munich (HMTM). AICA aims to equip students from HM and HMTM with the necessary knowledge and skills to comprehend and impact AI development in the creative and cultural industries. AICA operates digitally and physically, fostering knowledge exchange, practical collaboration and cultivating a network community. At its core, AICA is a teaching curriculum that unfolds in two stages. First, students can enroll in a tech crash course to gain theoretical and practical insights into AI technologies, particularly their applications in arts and culture. Then, next semester, students can enroll in a project workshop to apply this knowledge within multidisciplinary teams and develop their projects in AI in culture and arts. In addition, AICA is also a digital community and platform: it should promote student networking by establishing and moderating a digital community through a teaching and knowledge platform. A lecture series, documented on YouTube, is organized annually and is part of this endeavor.

This research investigates the effectiveness of the AICA curriculum, particularly the project workshop course, and reflects on the challenges and opportunities of the digitalization lab format. We are interested in the following research questions:

RQ1: Does the proposed teaching format foster students' AI literacy, particularly for students without a technological background (musicians, artists, designers, cultural managers)?

RQ2: Does the proposed teaching format foster students' self-efficacy in impacting the deployment of AI in culture and arts? Self-efficacy refers to “an individual's belief in his or her capacity to execute behaviors necessary to produce specific performance attainments” (Bandura, 1977).

RQ3: How do students, with or without a technological background, perceive the benefits of interdisciplinary teaching content and collaborations?

RQ4: How do students, with or without technological background, perceive the benefits of project-based learning?

To tackle these questions, we conducted a questionnaire study among 12 participants from the *project workshop* student cohort in 2023-2024. The questionnaire comprises two parts. First, Likert-scale and open-ended questions investigate the research questions above. Second, a multiple-choice question with a single correct answer measures students' AI literacy from a social and technical standpoint and further complements our investigation of RQ1. We used and refined the 16 questions of this objective measurement of AI literacy from Weber et al. [1]. This questionnaire's methodology and results are presented in sections 2 and 3 below. The paper proposes two complementary sets of questions, a subjective and objective questionnaire, to evaluate AI literacy in a project-based course and among students with a wide range of prior expertise.

2 METHODOLOGY

This section describes the methodology used to conduct the online questionnaire.

2.1 Participants

We administered the online questionnaire on the last day of the project workshop course and emailed a reminder. 12 students responded the questionnaire (M=6, F=4, Non-binary=1, Prefer not to say=1). Seven participants were between 18 and 24 years old, four were between 25 and 34, and 1 was between 35 and 44. Students' primary field of study covers four categories: Computer science (N=4), Cultural management and communication (N=4), Music or composition (N=3), and Design (N=4). In the demographic information, we also asked participants about their prior experience with programming. Five participants had no experience; 4 considered themselves beginner, two as intermediate, and only one as advanced.

2.2 Questionnaire design

The questionnaire is structured in three sections: 1) the self-reported benefits from the class, and 2) the objective AI literacy measurement, 3) and the demographics presented above.

The questions of the first sections were designed to address the digitalization lab objectives and the research questions reported in the introduction. The questions are presented in Appendix A. The questions of the objective AI literacy measurement were taken from Weber et al.[1] and reformulated. In line with previous work [2], we considered that AI literacy covers both technical and social topics and differs depending on the role of the stakeholder: a user, a creator, or an evaluator [3]. For this reason, the 16 questions Weber et al. [1] proposed were split into four categories, each including four questions: 1) Social and user-centered, 2) Social and creator/evaluator centered, 3) Technical and user-centered, and 4) Technical and creator/evaluator centered. In this publication, we simplified this taxonomy only to consider eight technical questions and eight social questions about AI. We also reformulated some of the questions for clarity. The final set of questions is listed in Appendix B.

3 RESULTS

This section introduces the questionnaire results, and its structure echoes the four research questions presented in the introduction.

3.1 AI literacy

In this subsection, we present the results on AI literacy that were gained from the questionnaire.

3.1.1 Students' AI literacy does not seem to be related to their primary field of study

Figure 1 shows students' scores to the objective AI literacy test listed in Appendix B (over 16 points).

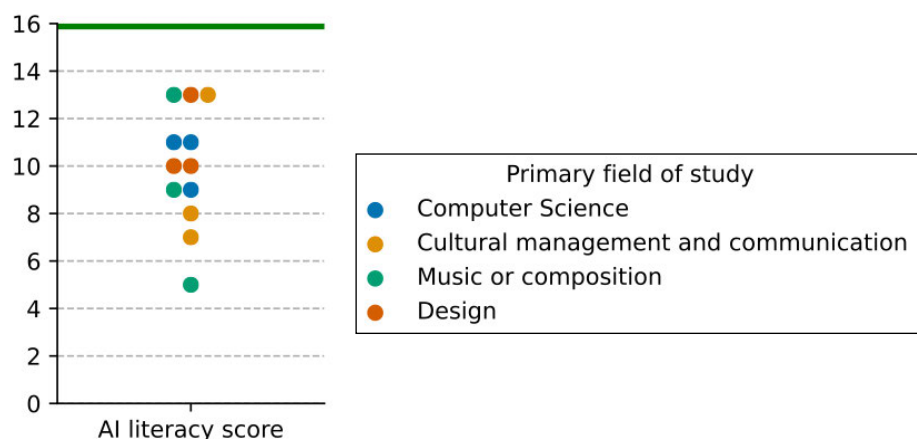


Figure 1. Students' objective AI literacy test scores. The color of each marker indicates the student's primary field of expertise. The horizontal green line indicates the best score attainable in the test.

Figure 1. a suggests that students' AI literacy seems independent of their primary field of study: musicians, designers, or students in cultural management can score better than computer scientists.

3.1.2 Students' AI literacy might be more technical than socially-oriented

Figure 2 shows the same scores split according to the technical and social questions about AI literacy (8 points each).

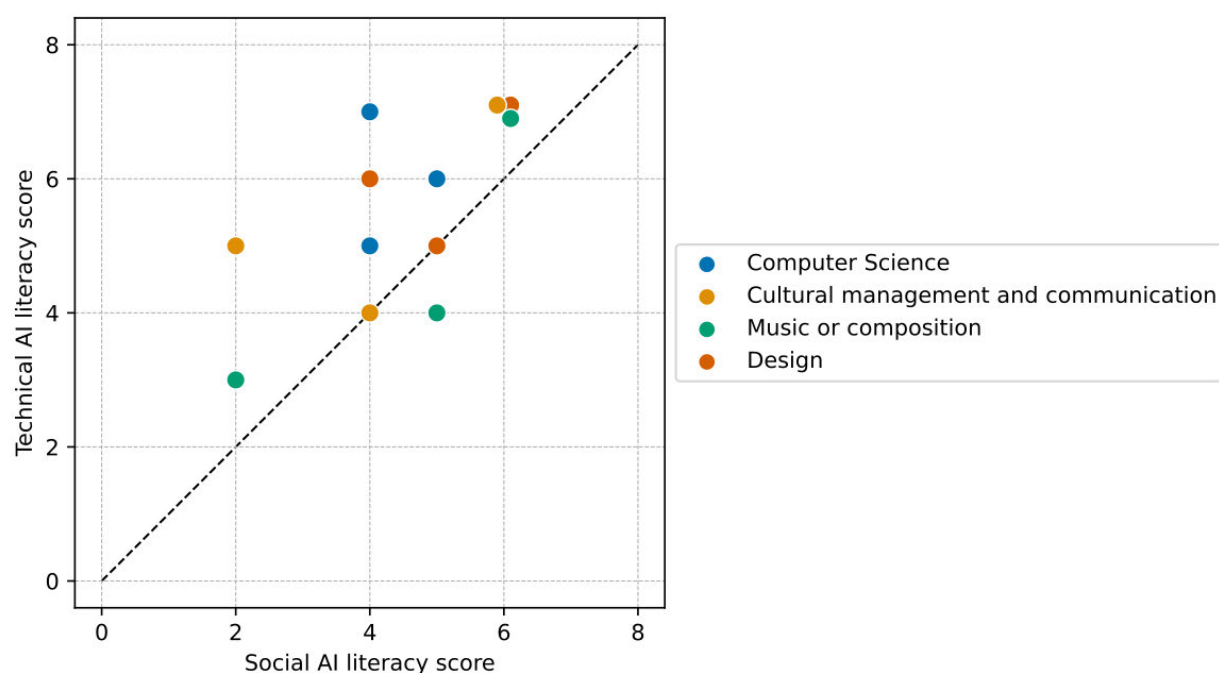


Figure 2. Students' social AI literacy score (x-axis) according to students' technical AI literacy score (y-axis), categorized by primary field of expertise.

The results of Figure 2 suggest that, overall, students might have better technical than social AI literacy after the course, as most obtained better scores on the eight technical questions than on the eight social questions. However, we acknowledge that the difficulty level might not be equal between the technical and social-related questions. In particular, the correct response rate largely varies across questions. For instance, the hardest question was question n°5 about current AI regulation (0/12 correct answer), while the easiest question was question n°3 about AI research (11/12 correct answer). This exploratory

research is the first endeavor to assess this objective of AI literacy measurement empirically. A promising research direction would be administering this questionnaire to a large pool of participants to calibrate the questions' difficulty and get a baseline for the populations' AI literacy.

3.1.3 Students perceived the benefit of the project workshop on their literacy about AI

Figure 3 shows the responses to Likert-scale question n°2 of the self-reported questionnaire.

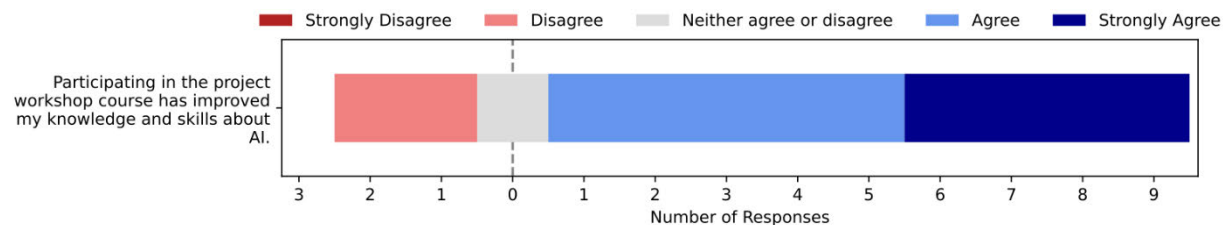


Figure 3. Students' response to Likert-scale question n°2 of the questionnaire: "Participating in the project workshop course has improved my knowledge and skills about AI."

Figure 3 suggests that most students agreed that participating in the project workshop course has improved their knowledge and skills about AI. Since students have different backgrounds and expertise regarding computer science and AI, one might wonder if the self-reported benefit of the teaching format is impacted by prior expertise. Figure 4 shows the same responses as in Figure 3, but the color of the points indicates the primary field of expertise. The figure does not suggest any relation to self-reported benefits on AI literacy regarding students' prior expertise.

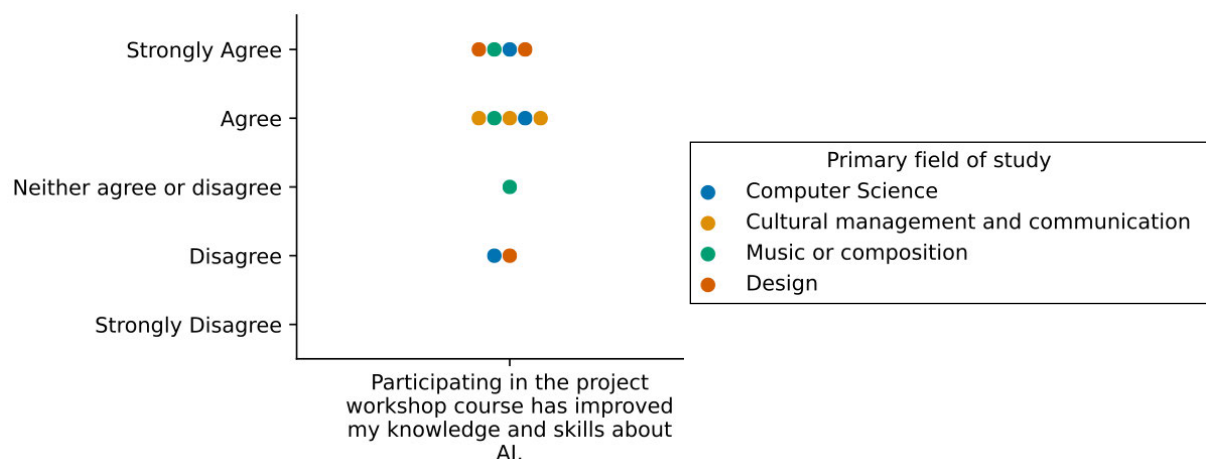


Figure 4. Students' response to Likert-scale question n°2 of the questionnaire: "Participating in the project workshop course has improved my knowledge and skills about AI". The color of each marker indicates the student's primary field of expertise.

Overall, these results attest to the usefulness of this teaching format, regardless of students' primary field of study. However, we still lack deeper insights on why the course was perceived as beneficial: was it because of the project-based format, the multidisciplinary approach, or was the narrow scope of application (culture and arts) useful to expand their literacy about AI?

3.2 Self-efficacy in impacting the deployment of AI in culture and arts

One of the goals of the Digitalization Lab is to involve non-AI experts in the development of AI, particularly when the technology is likely to affect their fields. In our case, non-AI experts include students from music, cultural management, and design. This section presents the results related to students' willingness to continue their efforts at the intersection of AI, culture, and arts. In psychology, self-efficacy "is an individual's belief in their capacity to act in the ways necessary to reach specific goals" [4]. Individuals' beliefs about their ability to influence situations impact their competence in facing challenges and the decisions they are likely to make.

Figure 5 shows the results of a Likert-scale question asking if students "want to continue developing their knowledge and skills in AI after this course".

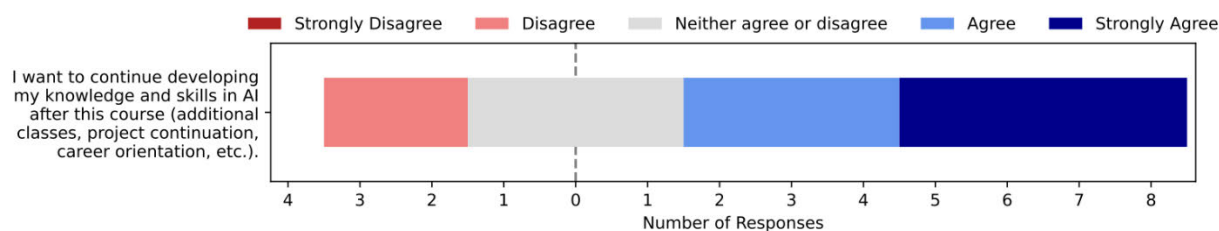


Figure 5. Students' response to Likert-scale question n°3 of the questionnaire: "I want to continue developing my knowledge and skills in AI after this course."

A large portion of students (3/12 agreed and 4/12 strongly agreed) agreed that they want to continue developing their knowledge and skills in AI after the course through additional classes, project continuation, career orientation, or other. Such a result is also a mark of success for the AICA project as it aligns with the objective of including diverse profiles, including non-technical profiles, in digitalization issues.

If students want to pursue their efforts at the intersection of AI and culture/arts, one might wonder if they also feel capable of doing so. We asked students if they "feel capable of applying AI concepts and techniques to impact the cultural and creative fields if they choose to." We also asked what their answers would have been before the project workshop course.

Figure 6 shows the results of two Likert-scale questions asking if students feel capable of applying AI concepts and techniques to impact the cultural and creative fields if they choose to.

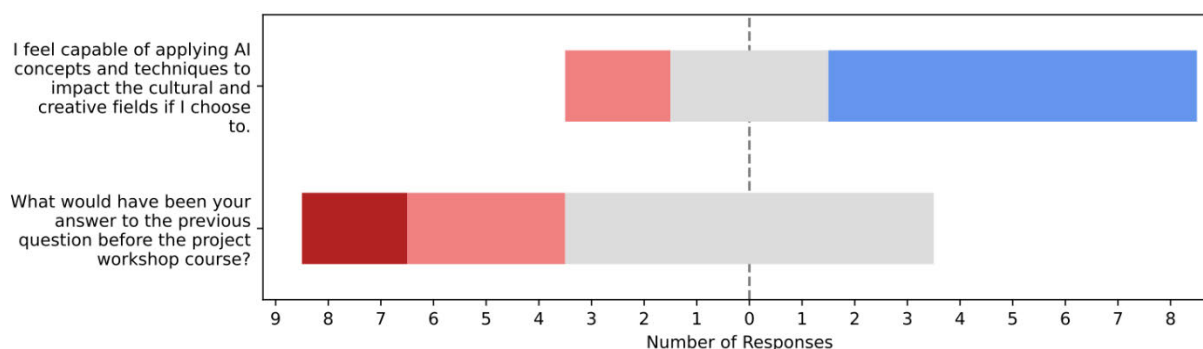


Figure 6. Students' response to Likert-scale question n°4 and 5 of the questionnaire: "I feel capable of applying AI concepts and techniques to impact the cultural and creative fields if I choose to" as well as a retrospective question asking "what would have been your answer to the previous question before the project workshop course?".

As shown in Figure 6, the majority of the participants agreed that they feel capable of applying AI concepts and techniques to impact the cultural and creative fields, especially compared to before the course. Of course, we acknowledge that retrospectively asking participants about their self-efficacy before the class presents biases, and we recommend designing a pre and post-questionnaire in future replications of this questionnaire. Once more, participants' responses at the individual level do not suggest any relationship to participants' prior expertise.

These results are overall promising for the utility of the AICA digitalization college as they suggest that course participants want and feel capable of and deeper investigations (e.g., qualitative study) are needed to understand students' self-efficacy to impact AI development in culture and arts.

3.3 Interdisciplinarity and project-based approach

The *project workshop* course is characterized by two key features: 1) students collaborate in interdisciplinary teams, 2) the course revolves around a project that is carried out and presented publicly at the end of the term. We asked students if they perceived specific benefits from these two characteristics and we report the results in the subsections below.

3.3.1 Students perceived the benefit of interdisciplinarity

Figure 7 shows the results of the Likert-scale questions asking if students "benefited from the interdisciplinary nature of the course." Most students agreed with that statement (6/12 agreed and 4/12

strongly agreed), suggesting that the course rationale to mix students with a wide range of expertise is perceived as beneficial.

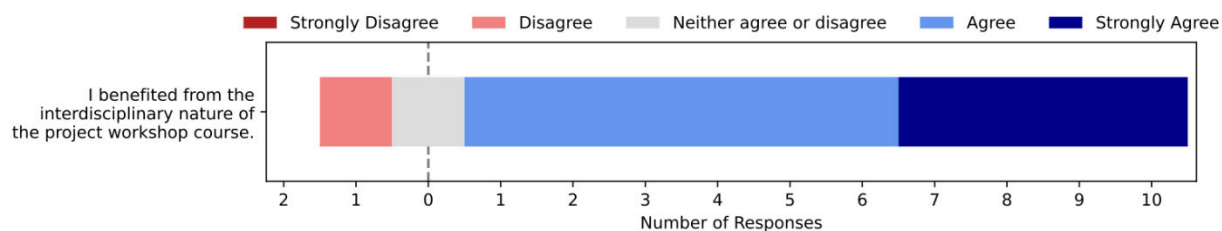


Figure 7. Students' response to Likert-scale question n°6 of the questionnaire: "I benefited from the interdisciplinary nature of the course"

Furthermore, we collected open-ended responses from the follow-up question asking to explain their response to question n°6. If the responses further emphasize the unique opportunity to bridge fields: "Without this course, I would probably never have had contact with computer scientists" (P3), the answers also shed light on the potential improvement of the course for the next iterations. Notably, the student who disagreed mentioned that "there were too few people with a technical background. The interdisciplinary nature wasn't as engaging as I hoped" (P6), suggesting that we should recruit more computer scientists to assist the technical implementation of the projects.

3.3.2 Students perceived the benefit of the project-based format

Figure 8 depicts the results of the Likert-scale questions asking students if "the project-based format of the course specifically helped them to develop their knowledge and skills about AI". Most students agreed with that statement (7/12 agreed and 1/12 strongly agreed), suggesting that the project-based format of the course is perceived as a good way to foster students' literacy about AI.

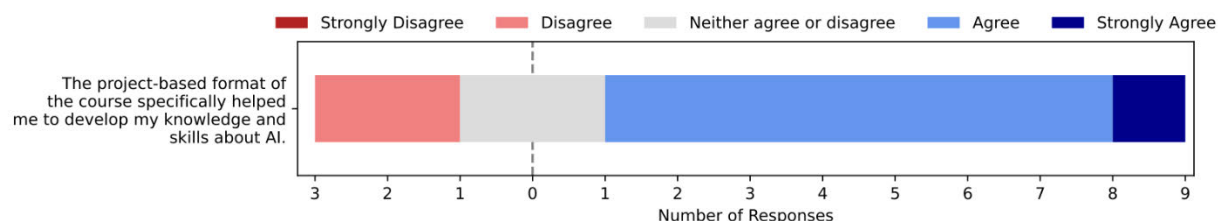


Figure 8. Students' response to Likert-scale question n°8 of the questionnaire: "The project-based format of the course specifically helped me to develop my knowledge and skills about AI"

Furthermore, the open-ended responses from the follow-up question mitigates that result and suggest to diversify the content of the class. For instance, participant 6 said that "the project was pushed too aggressively: there was no time to decide for oneself, what should be preferred. The AI skills were too specific for each group". Participant 8 suggested that "more lectures and small activities can help". Once more, participants' responses at the individual level do not suggest any relationship between perceived benefits of the interdisciplinary or project-based format and prior expertise.

4 CONCLUSIONS

This exploratory study details the design and conduct of a questionnaire study involving 12 students from a *project workshop* course at the BIDT digitalization labs on AI in culture and arts. The questionnaire assesses both an objective measurement of AI literacy from social and technical perspectives taken from Weber et al. [1], as well as the self-reported benefits of the course. The results suggest that students' objective and self-reported AI literacy levels do not depend on their primary field of expertise, which ranges across computer science, music, cultural management, and design. Furthermore, the findings suggest that students perceive the course—particularly its interdisciplinary and project-based aspects—as beneficial for their literacy and self-efficacy in learning about AI to culture and the arts. Importantly, this work not only provides the first empirical assessment of an existing objective measure of AI literacy [1], but also supplements it with the subjective evaluation we designed. Overall, this work offers a ready-to-use evaluation kit for multidisciplinary, project-based courses on AI.

ACKNOWLEDGEMENTS

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APPENDIX A. QUESTIONS ABOUT SELF-REPORTED BENEFITS FROM THE COURSE

- 1 What was your initial motivation to participate in this project workshop course? Please explain
- 2 Participating in the project workshop course has improved my knowledge and skills about AI.
Strongly disagree | Disagree | Neither agree or disagree | Agree | Strongly agree
- 3 I want to continue developing my knowledge and skills in AI after this course (additional classes, project continuation, career orientation, etc.).
Strongly disagree | Disagree | Neither agree or disagree | Agree | Strongly agree
- 4 I feel capable of applying AI concepts and techniques to impact the cultural and creative fields if I choose to.
Strongly disagree | Disagree | Neither agree or disagree | Agree | Strongly agree
- 5 What would have been your answer to the previous question before the project workshop course?
Strongly disagree | Disagree | Neither agree or disagree | Agree | Strongly agree
- 6 I benefited from the interdisciplinary nature of the project workshop course.
Strongly disagree | Disagree | Neither agree or disagree | Agree | Strongly agree
- 7 Please explain your previous answer.
- 8 The project-based format of the course specifically helped me to develop my knowledge and skills about AI.
Strongly disagree | Disagree | Neither agree or disagree | Agree | Strongly agree
- 9 Please explain your previous answer.

APPENDIX B. QUESTIONS ABOUT SELF-REPORTED BENEFITS FROM THE COURSE

<i>Item (n=16)</i>	<i>Answer Options (Correct answers in italics - randomized order)</i>
Social AI literacy	
1) The term artificial intelligence was first coined in...	A) the 1950s. B) the 1880s. C) the 1980s. D) ... the 2000s.
2) How are artificial and human intelligence related?	A) AI algorithms and humans demonstrate different cognitive faculties. B) AI algorithms are not biased because it learns from large and diverse data. C) AI is a faster simulation of the human brain. D) Only AI demonstrate biased behavior.
3) Artificial intelligence research...	A) has mainly been driven by machine learning advancements since the mid 2020s. B) is mainly interested in genetic algorithms. C) has not made significant theoretical progresses since the 1960s D) is mainly conducted by mathematicians.
4) What is a current risk of AI technology for humans?	A) The spread of fake and unattributable videos. B) Voice assistant hacks classified governmental documents C) Text generators might prevent the younger generation from learning how to write D) Autonomous artificial intelligence can disrupt the internet.
5) What is not an objective of current AI regulation?	A) Enforcing a 'no-bias policy 'to ban all potential biases that can arise from AI. B) Ensuring that AI systems placed on the market are safe and respect existing law on fundamental rights. C) Facilitating the development of a market for lawful, safe and trustworthy AI. D) Ensuring legal certainty to facilitate investment and innovation in AI.
6) Which is a major regulation that has been passed specifically for artificial intelligence?	A) European Commission's Act for Artificial Intelligence B) European Regulation for Responsible AI C) United Nations 'Framework for the Ethical Use of AI D) American Regulation on the Usage of AI
7) What is a possible consequence of the use of artificial intelligence at work?	A) The automation of tasks previously performed by workers. B) The debiasing of human literacy. C) The take-over of strategic decisions. D) Sending unwanted spams.
8) Key ethical issues surrounding artificial intelligence include...	A) bias and transparency. B) genetic algorithms and simulated annealing. C) predictions and overfitting. D) cold start problem and sunk cost fallacy.
Technical AI literacy	
9) What is the main distinction between supervised and unsupervised learning?	A) Supervised learning uses labeled datasets. B) Unsupervised learning precedes supervised learning. C) Unsupervised learning can occur at any moment in time. D) Supervised learning is performed by experts.
10) Which of the following statements is true?	A) Machine learning is part of artificial intelligence. B) Artificial intelligence is part of machine learning. C) Machine learning and artificial intelligence are mutually exclusive. D) Artificial intelligence and machine learning are identical.
11) What is the typical application for which artificial intelligence is generally better than classical computer programs?	A) Image recognition. B) Statistics calculation. C) Undefined processes. D) Hardware space analysis.
12) Running the same request with the same data on the same AI...	A) can give different results. B) triple the computing time. C) never give different results. D) increase the computing speed.
13) What is the main distinction between decision trees and support vector machines?	A) Decision trees are more interpretable. B) Decision trees are less interpretable. C) Decision trees give more complex predictions. D) Decision trees are trained faster.
14) What is a typical split of testing and training data for development purposes?	A) 80% training and 20% testing. B) 40% training, 40% testing, and 20% test and training altogether. C) 95% testing and 4% training. D) The split ratio does not matter.
15) What is not a required step in the development of a machine learning system?	A) Benchmarking. B) Data collection. C) Model choice. D) Training/ learning.
16) What is not part of an artificial neural network?	A) User layer. B) Input layer. C) Hidden layer. D) Output layer.